UPSC EPFO · APFC RECRUITMENT TEST · FOUNDATION LAYER · MODULE 9 · CHAPTER F9.2

Physics II — Optics, Heat, Sound, Electricity and Modern Physics

Chapter F9.2 — Containing the Two Optics Items the 2023 Paper Actually Asked

Module	**Foundation Layer — Module 9 — General Science & Computer Applications**
Chapter	**Chapter F9.2**
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NOTE · A RECALL LAYER, LIKE F9.1

Same shape as F9.1 and for the same reason: this is revision, not instruction. Tables, formulas, traps, practice.

Section 1 is the section to read properly. Two of the ten General Science questions in the 2023 paper were optics — the **power of a combination of lenses** and a **concave mirror with the object at infinity** (PYQ-DERIVED, Blueprint §4.4). Both are answered by material in §1.2 and §1.3.

HIGH YIELD · WHAT THE 2023 SCIENCE BLOCK ACTUALLY CONTAINED

Worth restating, because it shapes how you should read this chapter. The ten General Science questions in 2023 were:

Item	Where it is covered
Combined power of lenses	**F9.2 §1.3**
Concave mirror, object at infinity	**F9.2 §1.2**
Inelastic collision	F9.1 §3.2
Uniform acceleration of a train	F9.1 §2.2
Effluents and pH	F9.3
Soaps and detergents	F9.3
An enthalpy reaction	F9.3
Layers of the eyeball	F9.4
Banyan prop roots	F9.4
Ocean acidification	F9.4

Four of ten were mechanics and optics. That is the single strongest argument for the recall-layer approach: you are not learning this, you are reactivating it.

Table of Contents

0. How This Chapter Works

STUDY SESSION · CHAPTER F9.2 — FIVE SHORT SESSIONS

1. **Session 1** — Optics: mirrors, lenses, power, and light phenomena (§1) ★★★★★
2. **Session 2** — Heat and thermal behaviour (§2) ★★★★☆
3. **Session 3** — Sound (§3) ★★★☆☆
4. **Session 4** — Electricity and magnetism (§4) ★★★★★
5. **Session 5** — Modern physics and the spectrum (§5) ★★★★☆

Then 30 graded questions in §6. Budget **two days**, weighted towards §1 and §4.

1. Session 1 — Optics ★★★★★

1.1 Reflection and the two formulas

	Mirror	Lens
Formula	**1/v + 1/u = 1/f**	**1/v - 1/u = 1/f**
Magnification	m = **-v/u**	m = **v/u**
Converging type	**Concave**	**Convex**
Diverging type	**Convex**	**Concave**

Note the sign difference between the two formulas - a plus for mirrors, a minus for lenses

A **plane mirror** gives an image that is **virtual, erect, of the same size**, **laterally inverted**, and **as far behind the mirror as the object is in front**.

1.2 Spherical mirrors — the image table

MUST MASTER · CONCAVE MIRROR, OBJECT AT INFINITY

This is a 2023 question. For a **concave mirror with the object at infinity**, the image is formed **at the focus**, and it is **real, inverted and highly diminished** — effectively a point.

The full table, which is worth having complete because any row can be asked:

Object position	Image position	Nature
At infinity	**At the focus F**	Real, inverted, **highly diminished**
Beyond C	Between F and C	Real, inverted, diminished
At C	At C	Real, inverted, **same size**
Between C and F	Beyond C	Real, inverted, **enlarged**
At F	**At infinity**	Real, inverted, highly enlarged
Between F and pole	**Behind the mirror**	**Virtual, erect, enlarged**

Note the symmetry: object at infinity gives an image at F, and object at F gives an image at infinity. The two rows are each other's distractors.

A **convex mirror** is simpler and is asked as a single fact: it **always** forms a **virtual, erect and diminished** image, whatever the object position. That is exactly why it is used as a **rear-view mirror** — a wide field of view.

Uses: concave — shaving mirror, dentist's mirror, **headlight reflector**, solar furnace. Convex — **rear-view mirror**, street-light reflector.

1.3 Lens power — the other 2023 item

MUST MASTER · POWER, DIOPTRES AND COMBINATION

> **P = 1/f**, where f is the focal length **in metres**. The unit is the **dioptre (D)**.

Two consequences that between them answer the 2023 question:

1. **A convex (converging) lens has POSITIVE power; a concave (diverging) lens has NEGATIVE power.** So a lens of power -2 D is concave with a focal length of -0.5 m
2. **For thin lenses placed in contact, powers ADD: $P = P_1 + P_2 + \dots$**

So +2 D combined with +3 D gives **+5 D**. And +5 D combined with -2 D gives +3 D.

The trap is to add **focal lengths** instead of powers. Focal lengths combine as $1/f = 1/f_1 + 1/f_2$, which is why converting to power first is always the faster route.

1.4 The light phenomena that get asked

Phenomenon	Condition	Everyday example
Refraction	Light passes between media of different optical density	A pencil appearing bent in water
Total internal reflection	**Denser to rarer** medium, at **more than the critical angle**	**Optical fibre**, mirage, sparkle of a diamond
Dispersion	White light splits by wavelength through a prism	Rainbow
Scattering	Light deflected by particles; shorter wavelengths scatter more	**Blue sky**, **red sunrise and sunset**
Colour of an object	The object **reflects** that colour and absorbs the rest	A red book reflects red

COMMON UPSC TRAP · WHY THE SKY IS BLUE AND THE SUNSET RED — ONE CAUSE

Both are **scattering**, not reflection and not refraction.

Shorter wavelengths scatter far more strongly, so blue light is scattered across the sky, which is what you see when you look away from the Sun. At sunrise and sunset the light travels a much longer path through the atmosphere, so the blue has been scattered *out* of the beam by the time it reaches you, leaving the red.

One cause, two opposite-looking results. Options offering reflection or refraction for either are wrong.

1.5 Defects of vision

Defect	Problem	Corrected by
Myopia — short-sightedness	Image forms **in front of** the retina	**Concave** (diverging) lens
Hypermetropia — long-sightedness	Image forms **behind** the retina	**Convex** (converging) lens
Presbyopia	Loss of accommodation with age	**Bifocal** lens
Astigmatism	Unequal curvature of the cornea	**Cylindrical** lens

CHECK YOURSELF · SESSION 1

1. Give the mirror and lens formulas. What differs between them?
2. Concave mirror, object at infinity — where is the image and what is its nature?
3. What image does a convex mirror always form, and what is it used for?
4. Define the power of a lens and its unit. How do powers combine in contact?
5. Two lenses of +4 D and -1 D are in contact. What is the combined power and focal length?
6. Why is the sky blue and the sunset red? Which single phenomenon explains both?
7. Myopia and hypermetropia — which lens corrects each?

2. Session 2 — Heat ★★★★☆

2.1 Scales and transfer

- **K = °C + 273.15.** °F = (9/5)°C + 32. The two scales read the **same at -40°**
- Three modes of transfer: **conduction** (solids, no bulk movement), **convection** (fluids, bulk

movement), **radiation** (no medium needed — how the Sun's heat reaches us)

2.2 The numbers worth carrying

Quantity	Value
Specific heat capacity of water	about **4.2 kJ kg^{-1} K^{-1}** — unusually high
Latent heat of fusion of ice	about **336 kJ kg^{-1}** (80 cal g^{-1})
Latent heat of vaporisation of water	about **2,260 kJ kg^{-1}** (540 cal g^{-1})

Water's **high specific heat** is why it is used as a coolant, and why coastal climates are moderate.

MUST MASTER · THE ANOMALOUS EXPANSION OF WATER

Water has its maximum density at 4 °C. Below that temperature it *expands* on cooling, which is anomalous.

The consequence is the examinable part: **a lake freezes from the surface downwards.** As the surface water cools it becomes denser and sinks — but only until 4 °C. Below that it becomes *less* dense, so it stays on top, freezes there, and the ice then insulates the water beneath. That is why aquatic life survives a winter.

Note also that **ice is less dense than water**, which is why it floats — the reason an option saying "ice is denser than water" is wrong.

2.3 The laws of thermodynamics, at recognition level

Law	Content
Zeroth	Bodies in thermal equilibrium with a third are in equilibrium with each other — defines temperature
First	Energy is conserved: $\Delta Q = \Delta U + W$
Second	Heat cannot flow spontaneously from a colder to a hotter body; entropy of an isolated system increases
Third	Entropy approaches a constant as temperature approaches absolute zero

CHECK YOURSELF · SESSION 2

1. Convert 37 °C to kelvin and to fahrenheit
2. Name the three modes of heat transfer and which needs no medium
3. Give the latent heats of fusion of ice and vaporisation of water
4. At what temperature is water densest? Why does a lake freeze from the top?
5. Which law of thermodynamics is energy conservation? Which introduces entropy?

3. Session 3 — Sound ★★★☆☆

Property	Fact
Nature	A **longitudinal mechanical** wave
Medium	**Required** — sound **cannot travel through a vacuum**
Speed in air	about **343 m s^{-1}** at 20 °C
Speed by medium	**Fastest in solids**, slower in liquids, **slowest in gases**
Pitch	Depends on **frequency**
Loudness	Depends on **amplitude**
Quality or timbre	Depends on waveform

Frequency ranges: infrasonic below 20 Hz · **audible 20 Hz to 20,000 Hz** · **ultrasonic** above 20 kHz.

Applications: echo and **SONAR** (depth sounding), ultrasonography, reverberation control, the **Doppler effect** (apparent change in frequency due to relative motion).

COMMON UPSC TRAP · PITCH VERSUS LOUDNESS

Pitch depends on frequency. Loudness depends on amplitude. They are routinely swapped in options, and the swap is easy to miss because both words feel like "how the sound seems".

A shrill sound has a **high frequency**; a booming sound has a **large amplitude**. Sound also travels **fastest in solids** — the intuitive answer is air, because that is where we hear it, and that is exactly why the question works.

CHECK YOURSELF · SESSION 3

1. What kind of wave is sound? Can it travel in a vacuum?
2. In which medium is sound fastest, and in which slowest?
3. Pitch depends on what? Loudness on what?
4. Give the three frequency ranges

4. Session 4 — Electricity and Magnetism ★★★★★

4.1 The core relations

Relation	Formula
Current	**$I = Q/t$**, unit **ampere**
Ohm's law	**$V = IR$**
Resistance of a conductor	$R = \rho L/A$ — proportional to length, inversely to area
Series combination	**$R = R_1 + R_2 + \ldots$** — greater than the largest
Parallel combination	**$1/R = 1/R_1 + 1/R_2 + \ldots$ — less than the smallest**
Power	**$P = VI = I^2R = V^2/R$**
Heating effect	**$H = I^2Rt$** — Joule's law of heating

MUST MASTER · SERIES VERSUS PARALLEL, AND THE DIRECTION OF THE INEQUALITY

In series the equivalent resistance is greater than the largest individual resistance. In parallel it is LESS than the smallest.

The parallel half is what gets tested, because it is counter-intuitive — adding another path *reduces* total resistance, since current has more routes available. An option saying parallel resistance exceeds the largest component is wrong.

Practical corollary: household appliances are wired in **parallel**, so each gets the full supply voltage and can be switched independently.

4.2 Household electricity

- Three wires: **live** (usually red or brown), **neutral** (black or blue), **earth** (green or

yellow-green)

- **The fuse is connected in the LIVE wire**, so that a fault disconnects the supply from the appliance
- The **earth wire** is connected to the **metallic body** of an appliance, giving a low-resistance path

to earth and protecting the user from shock

- **India's supply: 220–240 V at 50 Hz**
- The commercial unit of energy is the **kilowatt-hour** — F9.1 §4.2

4.3 Magnetism and induction

Rule or device	What it does
Right-hand thumb rule	Direction of the magnetic field around a current-carrying conductor
Fleming's LEFT-hand rule	**Force** on a current-carrying conductor in a field — the **MOTOR** rule
Fleming's RIGHT-hand rule	Direction of **induced current** — the **GENERATOR** rule
Electric motor	Converts **electrical to mechanical** energy
Generator or dynamo	Converts **mechanical to electrical** energy
Transformer	Changes voltage; works on **mutual induction**; **AC only**

MUST MASTER · LEFT FOR MOTOR, RIGHT FOR GENERATOR — AND WHY A TRANSFORMER NEEDS AC

Two facts that are asked directly.

Fleming's left hand = motor (force produced from current). **Fleming's right hand = generator** (current produced from motion). The mnemonic: **left for a motor**, right for a generato**r** — or simply remember that the alphabet puts *generator* before *motor* while the hands go the other way.

A transformer works only on alternating current, because it depends on a **changing** magnetic flux to induce an emf in the secondary. Direct current produces a steady flux and therefore no induced emf once the current is established. So "a transformer can be used with DC" is wrong — and it is the reason transmission is done with AC.

CHECK YOURSELF · SESSION 4

1. State Ohm's law and the three expressions for electric power
2. Series and parallel — which gives a resistance less than the smallest component?
3. In which wire is the fuse connected, and why? What does the earth wire protect?
4. India's supply voltage and frequency?
5. Left-hand rule or right-hand rule — which is the motor rule?
6. Why does a transformer not work on direct current?

5. Session 5 — Modern Physics ★★★★★

5.1 The electromagnetic spectrum

In order of **increasing frequency** and therefore **decreasing wavelength**:

> **Radio waves » Microwaves » Infrared » Visible light » Ultraviolet » X-rays » Gamma rays**

Visible light runs from **violet** (highest frequency, shortest wavelength) to **red** (lowest frequency, longest wavelength) — remember VIBGYOR from violet to red.

> **MUST MASTER · THE ORDER, AND THE DIRECTION OF THE QUESTION**
>
> The sequence above is the whole of this topic, and the questions differ only in **which direction** they ask for.
>
> **Gamma rays have the highest frequency and the shortest wavelength; radio waves the lowest frequency and longest wavelength.** Read the stem for "increasing" or "decreasing", and for "frequency" or "wavelength" — four combinations, one sequence.

5.2 The nucleus

Term	Content
Radioactivity	Spontaneous emission of **alpha** (helium nucleus), **beta** (electron) and **gamma** (electromagnetic) radiation
Penetrating power	Gamma > beta > alpha. **Ionising** power is the reverse
Half-life	Time for half the nuclei to decay
Fission	A heavy nucleus splits. Used in **nuclear reactors** and the **atom bomb**
Fusion	Light nuclei combine. The energy source of the **Sun** and the **hydrogen bomb**

COMMON UPSC TRAP · FISSION VERSUS FUSION

Fission = splitting = reactors and the atom bomb. Fusion = joining = the Sun and the hydrogen bomb.

The words are nearly identical and the applications are routinely swapped. Anchor it on the Sun: the Sun **fuses** hydrogen into helium. Fusion requires enormous temperature and pressure, which is why we can build fission reactors but not yet commercial fusion ones.

Also at recognition level: the **photoelectric effect** and the **dual nature** of light; **X-rays** (discovered by Roentgen, used in imaging); **LASER**; and **semiconductors** as the basis of electronics.

CHECK YOURSELF · SESSION 5

1. Write the electromagnetic spectrum in order of increasing frequency
2. Which has the shortest wavelength? Which the longest?
3. Name the three radiations and rank their penetrating power
4. Fission or fusion — which powers the Sun? Which powers a nuclear reactor?

6. Graded Practice — 30 Questions ★★★★★

6.1 Level 1 — Recognition

Q1. The SI unit of the power of a lens is the (a) watt (b) metre (c) dioptre (d) joule

Q2. A convex lens is (a) converging (b) diverging (c) neither converging nor diverging (d) plane

Q3. For a concave mirror with the object at infinity, the image is formed (a) at the pole (b) at the centre of curvature (c) at infinity (d) at the focus

Q4. The mirror formula is (a) $1/v - 1/u = 1/f$ (b) $1/v + 1/u = 1/f$ (c) $v + u = f$ (d) $1/f = v/u$

Q5. Which mirror is used as a rear-view mirror in vehicles? (a) Concave (b) Plane (c) Convex (d) Cylindrical

Q6. Myopia is corrected by using (a) a convex lens (b) a cylindrical lens (c) a bifocal lens (d) a concave lens

Q7. Sound **cannot** travel through (a) a vacuum (b) air (c) water (d) steel

Q8. The audible range of frequency for a normal human ear is (a) 2 to 200 Hz (b) 20 to 20,000 Hz (c) 200 to 2,000 Hz (d) above 20 kHz

Q9. Ohm's law is expressed as (a) $V = I/R$ (b) $I = VR$ (c) $V = IR$ (d) $R = VI$

Q10. The frequency of the alternating current supply in India is (a) 60 Hz (b) 100 Hz (c) 25 Hz (d) 50 Hz

6.2 Level 2 — Understanding

Q11. Two thin lenses of power +2 D and +3 D are placed in contact. The power of the combination is (a) +5 D (b) +1 D (c) +6 D (d) +2.5 D

Q12. The focal length of a lens whose power is -2 D is (a) +0.5 m (b) -0.5 m (c) -2 m (d) +2 m

Q13. Total internal reflection occurs when light travels (a) from a rarer to a denser medium (b) along the normal (c) from a denser to a rarer medium at an angle greater than the critical angle (d) through a vacuum

Q14. The blue colour of the sky is due to (a) reflection of light (b) refraction of light (c) total internal reflection (d) scattering of light

Q15. Water has its maximum density at (a) 4 °C (b) 0 °C (c) 100 °C (d) -4 °C

Q16. In a series combination of resistors, the equivalent resistance is (a) less than the smallest resistance (b) the sum of the individual resistances (c) the reciprocal of the sum of the resistances (d) the average of the resistances

Q17. Electric power is given by (a) I^2R only (b) V/I (c) VI (d) V/R

Q18. A transformer works on the principle of (a) self-induction only (b) Ohm's law (c) the heating effect of current (d) mutual induction

Q19. A transformer can be used with (a) alternating current only (b) direct current only (c) both equally well (d) neither

Q20. Which one of the following has the **highest** frequency? (a) Radio waves (b) Gamma rays (c) Infrared radiation (d) Visible light

6.3 Level 3 — Recruitment Test standard

Q21. Which one of the following statements is **not** correct? (a) A concave mirror can form both real and virtual images (b) A convex mirror always forms a virtual, erect and diminished image (c) A concave lens can form a real image of a real object (d) The power of a lens is the reciprocal of its focal length expressed in metres

Q22. Consider the following statements:

1. In a domestic electric circuit the fuse is connected in the live wire.
2. The earth wire is a safety measure connected to the metallic body of an appliance.
3. In a parallel combination the equivalent resistance is greater than the largest individual resistance.

Which of the above statements are correct? (a) 1 and 3 only (b) 2 and 3 only (c) 1, 2 and 3 (d) 1 and 2 only

Q23. Match List-I with List-II and select the correct answer using the code given below:

List-I	List-II
A. Concave mirror	1. Headlight reflector
B. Convex mirror	2. Rear-view mirror
C. Convex lens	3. Correction of hypermetropia
D. Concave lens	4. Correction of myopia

(a) A-1 B-2 C-3 D-4 (b) A-2 B-1 C-4 D-3 (c) A-1 B-2 C-4 D-3 (d) A-3 B-4 C-1 D-2

Q24. The latent heat of fusion of ice is about (a) 2,260 kJ kg^{-1} (b) 336 kJ kg^{-1} (c) 4.2 kJ kg^{-1} (d) 100 kJ kg^{-1}

Q25. A lake freezes from the surface downwards because (a) ice is denser than water (b) water is a good conductor of heat (c) water has its maximum density at 4 °C, so colder water remains at the top and freezes there (d) the atmosphere warms the lower layers of the lake

Q26. Consider the following statements about sound:

1. Sound travels fastest in solids and slowest in gases.
2. The pitch of a sound depends on its amplitude.
3. Sound is a longitudinal mechanical wave.

Which of the above statements are correct? (a) 1 and 2 only (b) 2 and 3 only (c) 1, 2 and 3 (d) 1 and 3 only

Q27. Nuclear fusion is the source of the energy of (a) the Sun (b) a nuclear reactor (c) an atom bomb (d) a thermal power station

Q28. Fleming's left-hand rule gives the direction of the (a) induced current in a generator (b) force on a current-carrying conductor placed in a magnetic field (c) magnetic field around a straight current-carrying conductor (d) induced electromotive force

Q29. Arranged in order of **increasing frequency**, the correct sequence is (a) gamma rays, X-rays, ultraviolet, visible light (b) visible light, infrared, microwaves, radio waves (c) radio waves, infrared, visible light, ultraviolet (d) X-rays, visible light, infrared, radio waves

Q30. A body appears red in colour because it (a) absorbs red light (b) transmits all colours falling on it (c) emits red light of its own (d) reflects red light and absorbs the rest

7. Answers and Explanations

7.1 Level 1

Q	Ans	Explanation
1	c	The **dioptre**, the reciprocal of the focal length in metres
2	a	**Converging.** For lenses, convex converges; for *mirrors*, it is **concave** that converges — the reversal is worth fixing
3	d	**At the focus**, and the image is real, inverted and highly diminished. Option (c) is the mirror image of this case — object at F gives an image at infinity
4	b	**$1/v + 1/u = 1/f$** for a mirror. Option (a) is the **lens** formula
5	c	**Convex**, because it always gives a virtual, erect, diminished image and a wide field of view
6	d	A **concave** (diverging) lens. Hypermetropia needs a convex lens
7	a	A **vacuum** — sound is a mechanical wave and needs a material medium
8	b	**20 Hz to 20,000 Hz**
9	c	**$V = IR$**
10	d	**50 Hz**, at 220–240 V. 60 Hz is the North American standard

7.2 Level 2

Q	Ans	Explanation
11	a	**Powers add for thin lenses in contact**: +2 + 3 = **+5 D**. Adding focal lengths instead is the standard error
12	b	f = 1/P = 1/(-2) = **-0.5 m**. Negative power means a **concave** lens, and the sign must be carried through
13	c	**Denser to rarer, beyond the critical angle** — both conditions are required, which is why partial statements are wrong
14	d	**Scattering.** Shorter wavelengths scatter more strongly, so blue dominates the sky
15	a	**4 °C.** Below this water expands on cooling — the anomalous expansion
16	b	**The sum.** In parallel it is *less than the smallest*, which is the counter-intuitive half
17	c	**VI**, and equivalently I^2R and V^2/R. Option (a) says "only", which is what makes it wrong
18	d	**Mutual induction** between primary and secondary windings
19	a	**Alternating current only**, because a transformer needs a **changing** flux to induce an emf
20	b	**Gamma rays** — highest frequency and shortest wavelength in the spectrum

7.3 Level 3

Q	Ans	Explanation
21	c	A **concave lens cannot form a real image of a real object** — it is diverging and always gives a virtual, erect, diminished image. A concave *mirror* can form both real and virtual images, which is what (a) correctly states
22	d	**1 and 2 only.** Statement 3 is false: in **parallel** the equivalent resistance is **less than the smallest** component, because current has additional paths
23	a	**A-1 B-2 C-3 D-4** — concave mirror for a headlight reflector, convex mirror for rear view, convex lens for hypermetropia, concave lens for myopia
24	b	About **336 kJ kg^{-1}**. Option (a) is the latent heat of **vaporisation**, and the two are each other's distractors
25	c	Water is densest at **4 °C**, so water colder than that stays on top and freezes there, and the ice then insulates the water below. Option (a) reverses the density of ice and water
26	d	**1 and 3 only.** Statement 2 is false — **pitch depends on frequency**, not amplitude. Amplitude governs loudness
27	a	The **Sun**, which fuses hydrogen into helium. Reactors and the atom bomb use **fission**
28	b	The **force on a conductor** — the **motor** rule. The **right**-hand rule gives induced current, the generator case
29	c	**Radio, infrared, visible, ultraviolet** is the only option in increasing order of frequency. Option (a) is in *decreasing* order — read the stem's direction
30	d	It **reflects** red and absorbs the other wavelengths. A body does not emit its own colour unless it is a source

THINK LIKE UPSC · THE FOUR SHAPES IN THIS CHAPTER

1. **The mirror-lens reversal.** Q2, Q4 and Q21 all exploit it. **Concave mirror converges; convex lens converges.** Write that once on your wall sheet and check it every time
2. **Add powers, not focal lengths.** Q11 and Q12. Convert to dioptres first and the arithmetic becomes addition
3. **The direction of the stem.** Q29 gives you the right sequence in the wrong direction as a distractor. Underline "increasing" or "decreasing" before looking at the options
4. **The paired latent heats and paired velocities.** Q24 here, and escape versus orbital velocity in F9.1. Where two numbers belong to a matched pair, the examiner offers the other one

8. One-Minute Revision

ONE-MINUTE REVISION · CHAPTER F9.2

FORMULAS mirror **$1/v + 1/u = 1/f$** · lens **$1/v - 1/u = 1/f$** · **$P = 1/f$ (metres), unit dioptre** · **powers ADD in contact: $P = P_1 + P_2$**

CONVERGING concave MIRROR · **convex LENS** — the reversal that catches people

CONCAVE MIRROR, OBJECT AT INFINITY » IMAGE AT THE FOCUS, real, inverted, highly diminished · object at F » image at infinity

CONVEX MIRROR always **virtual, erect, diminished** » rear-view mirror

USES concave mirror: shaving, dentist, **headlight reflector**, solar furnace · convex mirror: rear view

SIGN OF POWER convex **+** · concave **-** · -2 D means f = **-0.5 m**

TOTAL INTERNAL REFLECTION denser to rarer, beyond the critical angle » optical fibre, mirage, diamond

SCATTERING explains **both** the blue sky **and** the red sunset · colour of an object = the light it **reflects**

VISION myopia » **concave** lens · hypermetropia » **convex** · presbyopia » bifocal · astigmatism » cylindrical

HEAT K = °C + 273.15 · -40° is the same in C and F · conduction, convection, **radiation needs no medium** · water specific heat **4.2 kJ/kg K** · **latent fusion of ice 336** · **latent vaporisation of water 2,260**

WATER DENSEST AT 4 °C » lakes freeze from the **top** down · **ice is LESS dense than water**

THERMODYNAMICS zeroth defines temperature · first is energy conservation · second introduces entropy

SOUND longitudinal, mechanical, **cannot travel in a vacuum** · **fastest in solids** · **pitch = frequency, loudness = amplitude** · infrasonic below 20 Hz · audible **20–20,000 Hz** · ultrasonic above 20 kHz · SONAR, Doppler

ELECTRICITY $I = Q/t$ · **$V = IR$** · $R = \rho L/A$ · **series = sum, greater than the largest** · **parallel = less than the SMALLEST** · **$P = VI = I^2R = V^2/R$** · **$H = I^2Rt$**

HOUSEHOLD fuse in the LIVE wire · **earth to the metallic body** · India **220–240 V, 50 Hz** · appliances wired in **parallel**

MAGNETISM Fleming LEFT = MOTOR (force) · **RIGHT = GENERATOR** (induced current) · motor: electrical to mechanical · generator: mechanical to electrical · **transformer = mutual induction, AC ONLY**

SPECTRUM, INCREASING FREQUENCY radio » microwave » infrared » visible » ultraviolet » X-ray » **gamma** · gamma shortest wavelength, radio longest · VIBGYOR violet to red

NUCLEUS alpha (helium nucleus) · beta (electron) · gamma (EM) · penetrating gamma > beta > alpha, ionising the reverse · **FISSION = splitting = reactors and atom bomb** · **FUSION = joining = the SUN and hydrogen bomb**

9. Five-Minute Wall Sheet

FIVE-MINUTE WALL SHEET · CHAPTER F9.2

CONCAVE MIRROR CONVERGES · CONVEX LENS CONVERGES — the reversal

MIRROR 1/v + 1/u · LENS 1/v - 1/u

P = 1/f(m), DIOPTRE · POWERS ADD · CONVEX + , CONCAVE -

OBJECT AT INFINITY » IMAGE AT FOCUS (concave mirror) · **CONVEX MIRROR ALWAYS VIRTUAL ERECT DIMINISHED**

MYOPIA » CONCAVE · HYPERMETROPIA » CONVEX

BLUE SKY AND RED SUNSET = SCATTERING (one cause)

TIR: DENSER TO RARER, BEYOND CRITICAL ANGLE

WATER DENSEST AT 4 °C · ICE FLOATS · LAKES FREEZE TOP DOWN

FUSION 336 · VAPORISATION 2,260 kJ/kg

SOUND: NO VACUUM · FASTEST IN SOLIDS · PITCH = FREQUENCY

SERIES = SUM · PARALLEL = LESS THAN THE SMALLEST

$P = VI = I^2R = V^2/R$ · FUSE IN LIVE WIRE · 220–240 V, 50 Hz

LEFT HAND = MOTOR · RIGHT HAND = GENERATOR · TRANSFORMER = AC ONLY

RADIO » MICROWAVE » INFRARED » VISIBLE » UV » X-RAY » GAMMA (increasing frequency)

FISSION = REACTOR · FUSION = SUN

ONE LINE FOR THE INTERVIEW A transformer works only on alternating current because induction requires a changing magnetic flux — which is the whole reason electrical power is transmitted as AC rather than DC.

ACTION · NEXT IN MODULE 9

Chapter F9.3 — Chemistry, which covers the remaining three 2023 science items: **effluents and pH**, **soaps and detergents**, and **an enthalpy reaction**. Then F9.4 Biology and Environment, and F9.5 to F9.7 on Computer Applications.

Before you move on, state without looking: which of concave and convex converges for a mirror and for a lens; where the image forms for a concave mirror with the object at infinity; how lens powers combine; which is less than the smallest, series or parallel; and which hand is the motor rule.